

PcpB (P42535) — Ancestral *para*-substituted-phenol hypothesis: focused curation report

Gene: pcpB (pentachlorophenol 4-monooxygenase), *Sphingobium chlorophenolicum* (NCBITaxon:46429) **UniProt:** P42535 (PCPB_SPHCR), 538 aa; EC 1.14.13.50; FAD cofactor; PheA/TfdB FAD monooxygenase family (Pfam PF01494 FAD_binding_3) **Focus type:** free_text — hypothesis slug `ancestral-phenol-pocket-alphaFold` **Hypothesis:** The ancestral substrate of PcpB is a naturally occurring *para*-substituted phenol, and the AlphaFold pocket + published SAR should narrow the candidate set.

Executive Judgment

Verdict: Partially supported (biochemically well-supported premise; structurally unresolved by direct measurement).

The *chemical* half of the hypothesis is strongly supported. PcpB is a *para*-specific phenol monooxygenase with documented promiscuity across *para*-substituted phenols (halo-, nitro-, amino-, cyano-), and its published structure–activity relationship (low phenolic pKa, hydrophobicity, an *ortho* substituent) plus the "recently assembled pathway" evolutionary context make it very plausible that the enzyme descends from one acting on a naturally occurring *para*-substituted phenol rather than on the man-made pesticide pentachlorophenol (PCP, introduced in the 1930s). An SAR-based ranking narrows the natural candidate set to **biogenic ortho-halogenated para-substituted phenols with free 3,5 positions** (2,4,6-tribromophenol, 2,4,6-trichlorophenol, 2,6-dichlorophenol, 2,4-dibromophenol), with nitrophenols as secondary candidates and methoxyphenols / plant alkyl-phenols essentially excluded on electronic or positional grounds.

The *structural* half — "measure the pocket in the AlphaFold model and read off which phenols fit its geometry" — **could not be delivered as a direct computation**, and I say so plainly. The AlphaFold model (AF-P42535-F1, v6) is high quality but contains **no FAD and no substrate**, and the nearest experimental structure (p-hydroxybenzoate hydroxylase, PHBH) is only ~20%

identical, so a cross-identity superposition to import the flavin/substrate position failed (least-squares fit on 3 atoms). Consequently, pocket volume and the substrate-to-C4a-hydroxyflavin distance were **not measured de novo**; the candidate ranking rests on the *published SAR + handbook physicochemistry*, not on a pocket-geometry measurement. The mechanistic uncoupling explanation is therefore an SAR/electronic argument, not a structural distance measurement.

Most important caveats: - The specific ancestral molecule is *unknowable* from current evidence; "a biogenic ortho-halophenol" is the best-constrained candidate *class*, not a proven identity. - The seed-provided mmCIF URL (...model_v4.cif) is retired; the live model is **v6**. All results here use v6. - Candidate scoring is a transparent, assumption-driven heuristic built from published SAR rules; it ranks plausibility, it does not prove turnover.

This is a **background/evolutionary insight**, not a function-assignment change. It should **not** alter the core experimentally-supported GO annotations (see GO Curation Implications).

Evidence Matrix

#	Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding
1	22482720 (Hlouchova et al. 2012, Biochemistry)	Direct assay + SAR	Supports (premise)	PcpB SAR; PCP is a poor/non- ancestral substrate	Binding/activity enhanced by low phenolic pKa, hydrophobicity, and an <i>ortho</i> substituent; kcat(PCP)=0.024/s "not well evolved" pathway "assembled ... by ... horizontal gene transfer"; uncoupling 0–100% ↑ by bulky 3/4/5 substituents, ↓ by ortho-Cl
2	10907421 (Zablotowicz et al. 1999)	Direct assay (recombinant)	Supports	Para- substituted- phenol substrate range	pcpB (in <i>E. coli</i>) active on p-nitrophenol, 4- nitrocatechol, 2,4- DNP, 4,6-dinitro-o- cresol; NOT 2,6- DNP, o-/m- nitrophenol, picric acid
3	UniProt P42535 (ECO:0000269)	PubMed:12169590, 22482720)	Database (expert, exp-backed)	Supports	Para-specific hydroxylation

#	Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding
4	23676275 (Yadid et al. 2013, PNAS)	Mechanistic/ evolutionary	Qualifies/ Supports	Why turnover is slow	Slow PcpB turnover (0.02/s) is maintained by selection so PcpD can sequester the toxic TCBQ intermediate
5	This work — AlphaFold DB AF-P42535-F1 v6; <code>parse_alphafold_confidence</code> ; <code>phenix.molprobity</code>	Computational (structural)	Supports (fold/ confidence)	Fold identity & model confidence	Mean pLDDT 91.5; FAD motifs res 16–45 (97.4) and 288–298 (97.3) very high; GxGxxG at 226; MolProbity 0.9 clashscore 2.04
6	This work — Foldseek webserver (search.foldseek.com, 3Di; pdb100 + afdb-swissprot)	Computational (structural neighbours)	Supports (family)	Nearest structural neighbours	All FAD-dependent aromatic/phenol monooxygenases & 20–28% id: 6-methylpretetramic 4-MO (Q3S8Q4), OxyS (4k2x), PgaE (2qa1), VibO (7yj0), PieE+FAD+substrate (6u0s)
7	This work — <code>phenix.superpose_pdbs</code> vs PHBH (PDB 1PBE, FAD + p-hydroxybenzoate)	Computational (negative)	Qualifies (limitation)	Transfer flavin/ substrate position to measure pocket	20.1% identity, LS fit on 3 atoms ⇒ degenerate; pocket geometry & substrate–C4a distance not

#	Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding
					measurable de novo
8	This work — SAR ranking (analysis/pcpB_candidate_ranking.csv)	Computational (heuristic)	Supports (narrows set)	Which natural phenols fit SAR	Top: 2,4,6-tribromophenol, 2,4,6-trichlorophenol, 2,6-dichlorophenol, 2,4-dibromophenol, nitrophenols secondary; methoxyphenols/ plant alkylphenols excluded; PCP bin best but couples worst
9	21903310; 26517387	Review/field data	Supports (biogenic availability)	Are top candidates naturally occurring?	2,4,6-tribromophenol is dominant natural marine product; brominated aromatics are biosynthesised by algae/sponges/ cyanobacteria

GO Curation Implications (leads — require curator verification)

This hypothesis is **evolutionary background**, not a function reassignment. It does **not** justify removing or weakening the experimentally-supported core annotations.

GO term	Aspect	Current basis	Recommended action (lead)
GO:0018677 pentachlorophenol monooxygenase activity	MF	IEA (UniProtKB- EC); experimentally backed (P 22482720 P 25238136)	Retain. Experimentally demonstrated; PCP is the validated in vitro/in vivo substrate regardless of evolutionary origin. Can be upgraded from IEA to EXP/IDA with (P 22482720)
GO:0071949 FAD binding	MF	IEA (InterPro)	Retain. Cofactor confirmed (P 22482720); FAD motifs at very high pLDDT. Upgradeable to IDA.
GO:0019338 pentachlorophenol catabolic process	BP	IEA (UniPathway)	Retain; rate-limiting step (P 22482720).
Consider adding — MF generalization	MF	Promiscuity data	Lead: the enzyme's demonstrated activity is broader than PCP. A parent/accompanying MF such as GO:0018685 "alkane 1-monooxygenase" -style is wrong; the correct broader parents are GO:0016709 (oxidoreductase, acting on paired donors, with incorporation of one O, NAD(P)H as one donor) and the family term. A more informative lead is annotating the para-nitrophenol / chlorophenol 4-monooxygenase activity shown in (P 10907421) if a suitable MF child exists. Verify ontology availability before use.
"Ancestral substrate = biogenic halophenol"	—	This analysis	Do NOT add as an annotation. It is a hypothesis about evolutionary history, not a current molecular function; at most a free-text note in the review.

Bottom line for curation: keep the PCP monooxygenase, FAD-binding, and PCP catabolism annotations; optionally record (non-core, background) that PCP is a recently-acquired xenobiotic substrate of a para-specific phenol hydroxylase whose promiscuity spans natural para-substituted phenols. Avoid "protein binding" — a specific MF (PCP 4-monooxygenase activity; FAD binding) is well supported.

Mechanistic Scope

Direct molecular function (what is tested): FAD-dependent, NADPH-consuming aromatic *para*-hydroxylation of phenols — a single-component Class A flavoprotein monooxygenase that forms a C4a-(hydro)peroxyflavin and hydroxylates the ring *para* to the phenolic –OH, replacing H, Cl, NO₂, NH₂ or CN. This is the gene product's primary, direct activity.

Downstream / not the direct activity: PCP detoxification, the assembled PCP catabolic pathway, growth on PCP, and the physiological H₂O₂/futile-cycle burden are pathway- and phenotype-level consequences. The "ancestral substrate" is an evolutionary inference about the enzyme's history and must be separated from its measured present-day activity.

Uncoupling (mechanistic question): The SAR gives a *substrate-electronic/steric* explanation — uncoupling rises with bulky substituents at ring positions 3/4/5 and falls with an ortho-chlorine (

P 22482720

PCP is fully substituted (3,4,5-Cl present), which is exactly the pattern that maximises uncoupling; the ring is plausibly held sub-optimally relative to the C4a-hydroxyflavin so the oxygenating intermediate decays to H₂O₂. **I could not convert this into a measured substrate-flavin distance** because the model lacks FAD/substrate and the homolog superposition was degenerate (Finding 7). So the uncoupling account is SAR-based inference, not a structural measurement.

Conflicts and Alternatives

1. Slow kcat is partly adaptive, not merely "unevolved"

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P 23676275

Selection maintains slow PcpB turnover so PcpD can sequester the highly toxic TCBQ

intermediate. This complicates the simple "PCP is a bad fit so the enzyme must be optimised for something else" narrative: some of the poor performance on PCP is *selected*, not just *ancestral mismatch*.

2. **Ortholog vs paralog / strain identity.** The broad para-phenol substrate data

(P 10907421)

come from *Sphingomonas* UG30 pcpB, a near-identical ortholog, not literally the ATCC 39723 P42535 protein. High identity makes transfer reasonable but should be verified.

3. **Natural availability ≠ ancestral exposure.** Biogenic tribromophenol is abundant in *marine* systems; *S. chlorophenicum* is a soil organism. The ancestral enzyme more likely met *terrestrial* microbial/fungal chlorophenols or nitrophenols. The candidate *class* (ortho-halo/para-substituted phenol) is robust; the specific molecule is not.

4. **Methoxyphenols dismissed on electronics, not sterics.** Guaiacol/syringol are abundant natural (lignin) phenols but have high pKa (~9.9); they fail the "low pKa" SAR rule, so their exclusion depends on the electronic rule holding — worth an explicit assay.

5. **Heuristic scoring risk.** My ranking collapses binding and coupling into simple additive terms; a different weighting could reorder nitrophenols vs dihalophenols. The qualitative tiers (halophenols/nitrophenols » methoxyphenols » alkylphenols) are more robust than the exact numbers.

Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
No experimental PcpB structure / no FAD-substrate complex	Foldseek + superposition; nearest exp. homolog ~20% id	Prevents direct pocket measurement and docking	Crystal/cryo-EM of PcpB + FAD + substrate analog; or AlphaFold3/Boltz with FAD+ligand
Pocket geometry not measured de novo	AlphaFold v6 has no ligand; superpose failed (3-atom fit)	The seed's "decisive analysis" is unmet	Ligand-aware modelling + cavity analysis (fpocket/CASTp) on a flavin-docked model
Actual ancestral substrate unknown	Literature + SAR ranking	Determines whether any specific GO/NTR note is warranted	Ancestral sequence reconstruction + resurrection kinetics across the TfdB/PheA clade
kcat/coupling for the predicted natural candidates	Only PCP/tetrachlorophenol/nitrophenols measured	Would test the ranking directly	Steady-state kcat + % uncoupling (H ₂ O ₂ /O ₂) for 2,4,6-tribromophenol, 2,4,6-trichlorophenol, 2,6-dichlorophenol
Soil vs marine source of natural halophenols	Marine biogenic data only	Ecological plausibility of ancestral exposure	Survey terrestrial microbial/fungal chloro-/nitro-phenol production

Discriminating Tests

- 1. Kinetic coupling screen (most decisive):** measure kcat, KM and % uncoupling (H₂O₂ production per NADPH) for the ranked natural candidates — predict high coupling + reasonable kcat for 2,4,6-trihalophenols and 2,6-dichlorophenol, high uncoupling for PCP, poor binding for guaiacol/syringol, and no turnover for p-cresol.
- 2. Ligand-aware structure prediction:** AlphaFold3/Boltz-1 of PcpB + FAD (+ candidate phenol), then cavity analysis (fpocket/CASTp) and measurement of the ring-C4 to flavin-C4a distance across substrates — the structural step this run could not complete.

3. **Ancestral sequence reconstruction (ASR):** reconstruct and resurrect ancestral nodes of the PcpB/TfdB/PheA clade; assay substrate profiles to read the ancestral preference directly.
4. **Active-site mutagenesis:** probe predicted pocket residues (e.g., the aromatic/hydrophobic residues lining the flavin cleft) for effects on uncoupling vs turnover.

Curation Leads (require curator verification)

- **Candidate references to cite in the review (verify snippets):**

- PMID 22482720 — "substrate binding and activity are enhanced by a low pK(a) for the phenolic proton, increased hydrophobicity, and the presence of a substituent ortho to the hydroxyl group of the phenol"; and "...increased by the presence of bulky substituents at position 3, 4, or 5 and decreased by the presence of a chlorine in the ortho position."
- PMID 10907421 — "oxidatively metabolized certain other p-substituted nitrophenols, i.e., p-nitrocatechol, 2,4-dinitrophenol (2,4-DNP), and 4,6-dinitrocresol..."; "...were also suitable substrates for the UG30 PCP-4-monooxygenase (pcpB gene expressed in Escherichia coli)."
- PMID 23676275 — "The toxicity of TCBQ may have exerted selective pressure to maintain slow turnover of PcpB (0.02 s⁻¹)."
- PMID 21903310 — "2,4,6-triBPh was dominant isomer in all cetaceans" (biogenic halophenol availability).
- **GO actions:** Retain GO:0018677 (MF), GO:0071949 (MF), GO:0019338 (BP); consider EXP/IDA upgrades from

P 22482720

Do **not** add any "ancestral/biogenic halophenol" GO term — keep as free-text background only.

- **Suggested questions for the curator:** (a) Should the review note that PCP is a *recently acquired xenobiotic* substrate of a para-specific phenol hydroxylase (promiscuity spans natural para-substituted phenols)? (b) Is a broader MF (para-substituted phenol 4-monooxygenase) desired alongside the PCP-specific term, given

P 10907421

- **Suggested experiments:** the kinetic coupling screen and ligand-aware modelling above.

Tools and versions actually invoked (provenance)

- **AlphaFold DB:** entry P42535; model **AF-P42535-F1 version 6** (model date 2025-08-01; AlphaFold Monomer v2.0 pipeline; global pLDDT 91.44). *Note:* the seed-cited `...model_v4.cif` URL returns NoSuchKey (retired); v6 used. Files: `model_v6.cif/.pdb`, `confidence_v6.json`, `PAE_v6.json`.
- **UniProt REST:** P42535.txt (entry v106, 2026-06-10).
- `parse_alphafold_confidence` (MCP): mean pLDDT 91.5; per-region values as reported.
- `phenix.molprobity` (Phenix): MolProbity 0.97; clashscore 2.04; Rama favored 98.13%.
- `phenix.superpose_pdbs` (Phenix): PcpB vs PDB **1PBE** (PHBH); 20.1% id; degenerate 3-atom LS fit (inconclusive).
- **Foldseek webserver** (search.foldseek.com, 3Di mode) vs **pdb100** and **afdb-swissprot**: neighbour list as reported (ran 2026-10-09).
- **Experimental references fetched (RCSB):** 1PBE, 2QA1 (PgaE), 6U0S (PieE).
- **PubMed** (MCP search_pubmed): PMIDs 22482720, 10907421, 23676275, 21903310, 26517387, 18452974.
- **Custom Python** (pandas/numpy/matplotlib): SAR candidate ranking → `analysis/pcpB_candidate_ranking.csv` + figures.

Candidate ranking (computed; top rows)

Phenol	Class	pKa	logP	ortho-sub	para OK	ortho-halogen	bulky 3/4/5	Binding fit	Coupling quality
2,4,6-tribromophenol	biogenic halophenol	6.10	4.13	Y	Y	Y	0	5.10	2
2,4,6-trichlorophenol	halophenol	6.23	3.69	Y	Y	Y	0	4.86	2
2,6-dichlorophenol	biogenic halophenol	6.79	2.75	Y	Y	Y	0	4.20	2
2,4-dibromophenol	biogenic halophenol	7.80	3.50	Y	Y	Y	0	4.00	2
4-nitrocatechol	nitrophenol	6.70	1.10	Y	Y	N	0	3.59	1
4-nitrophenol	nitrophenol	7.15	1.91	N	Y	N	0	2.19	1
guaiacol	methoxyphenol	9.93	1.32	Y	Y	N	0	2.06	1
pentachlorophenol	xenobiotic	4.70	5.12	Y	Y	Y	3	6.15	0
p-cresol	plant phenolic	10.26	1.94	N	N	N	0	0.00	1
tyrosol	plant phenolic	10.40	0.50	N	N	N	0	0.00	1

Binding fit and coupling quality are transparent heuristics from published SAR + handbook physicochemistry, not docking scores. PCP = best binder, worst coupler (mechanistic signature of its extensive uncoupling).